112 Autumn

EEIE30076

系統晶片設計 SOC Design

Lab. #2

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系級：機械所碩一

一、 Brief introduction about the overall system

學習兩種端口(AXI-Master/AXI-Stream)將資料從 PS(Processing System)傳送至 PL(Programable Logic)。

二、 What is observed & learned

- Differences between MAXI and Stream interface
  AXI-Master:

  傳輸須帶有地址，可傳至特定的地址

  可傳送指標/陣列參數

  輸入/輸出地址必須要是連續記憶體地址

  一條端口僅能一個讀+寫在一次的迴圈裡

  AXI-Stream:

  傳輸不帶有地址

  不能傳送指標陣列參數

  可無限制突發傳送(burst transfer)

- **Differences between csim and cosim**
  C Simulation:

  僅驗證(Validation)Code 是否可以通過 testbench 的測試集

  C Co-simulation:

  確認(Verification)Code 生成出來的電路是否符合延遲時間

三、 Screen dump

- Performance

```
================================================================
== Vitis HLS Report for 'fir_n11_maxi'
================================================================
* Date:           Tue Sep 26 20:08:58 2023

* Version:        2022.1 (Build 3526262 on Mon Apr 18 15:48:16 MDT 2022)
* Project:        Lab2_FIRN11MAXI
* Solution:       solution1 (Vivado IP Flow Target)
* Product family: zynq
* Target device:  xc7z020-clg400-1


================================================================
== Performance Estimates
================================================================
+ Timing:
    * Summary:
    +--------+---------+----------+------------+
    |  Clock |  Target | Estimated| Uncertainty|
    +--------+---------+----------+------------+
    |ap_clk  |  10.00 ns|  7.300 ns|     2.70 ns|
    +--------+---------+----------+------------+

+ Latency:
    * Summary:
    +---------+---------+----------+----------+-----+-----+---------+
    |  Latency (cycles) |  Latency (absolute) |  Interval | Pipeline|
    |   min   |   max   |    min   |    max   | min | max |   Type  |
    +---------+---------+----------+----------+-----+-----+---------+
    |        ?|        ?|         ?|         ?|    ?|    ?|       no|
    +---------+---------+----------+----------+-----+-----+---------+

    + Detail:
        * Instance:
        +----------------------------------------+-------------------------------+---------+---------+----------+----------+-----+-----+---------+
        |                                        |                               |  Latency (cycles) |  Latency (absolute) |  Interval | Pipeline|
        |                Instance                |             Module            |   min   |   max   |    min   |    max   | min | max |   Type  |
        +----------------------------------------+-------------------------------+---------+---------+----------+----------+-----+-----+---------+
        |grp_fir_n11_maxi_Pipeline_XFER_LOOP_fu_242  |fir_n11_maxi_Pipeline_XFER_LOOP  |        ?|        ?|         ?|         ?|    ?|    ?|       no|
        +----------------------------------------+-------------------------------+---------+---------+----------+----------+-----+-----+---------+

        * Loop:
        N/A
```

Performance (AXI-Master)

```
================================================================
== Performance Estimates
================================================================
+ Timing:
    * Summary:
    +--------+---------+----------+------------+
    |  Clock |  Target | Estimated| Uncertainty|
    +--------+---------+----------+------------+
    |ap_clk  | 10.00 ns|  6.923 ns|     2.70 ns|
    +--------+---------+----------+------------+

+ Latency:
    * Summary:
    +---------+---------+----------+----------+-----+-----+---------+
    |  Latency (cycles) |  Latency (absolute) |  Interval | Pipeline|
    |   min   |   max   |    min   |    max   | min | max |   Type  |
    +---------+---------+----------+----------+-----+-----+---------+
    |        ?|        ?|         ?|         ?|    ?|    ?|       no|
    +---------+---------+----------+----------+-----+-----+---------+

    + Detail:
        * Instance:
        +---------------------------------------+--------------------------------+---------+---------+----------+----------+-----+-----+---------+
        |                                       |                                |  Latency (cycles) |  Latency (absolute) |  Interval | Pipeline|
        |                Instance               |             Module             |   min   |   max   |    min   |    max   | min | max |   Type  |
        +---------------------------------------+--------------------------------+---------+---------+----------+----------+-----+-----+---------+
        |grp_fir_n11_strm_Pipeline_XFER_LOOP_fu_112  |fir_n11_strm_Pipeline_XFER_LOOP  |        ?|        ?|         ?|         ?|    ?|    ?|       no|
        +---------------------------------------+--------------------------------+---------+---------+----------+----------+-----+-----+---------+


        * Loop:
        N/A
```

Performance (AXI-Stream)

- Utilization

```
================================================================
== Utilization Estimates
================================================================
* Summary:
+-----------------+---------+-----+--------+-------+-----+
|       Name      | BRAM_18K| DSP |   FF   |  LUT  | URAM|
+-----------------+---------+-----+--------+-------+-----+
|DSP              |        -|    -|       -|      -|    -|
|Expression       |        -|    -|       0|     40|    -|
|FIFO             |        -|    -|       -|      -|    -|
|Instance         |        0|   33|    3806|   2838|    -|
|Memory           |        -|    -|       -|      -|    -|
|Multiplexer      |        -|    -|       -|    175|    -|
|Register         |        -|    -|     650|      -|    -|
+-----------------+---------+-----+--------+-------+-----+
|Total            |        0|   33|    4456|   3053|    0|
+-----------------+---------+-----+--------+-------+-----+
|Available        |      280|  220|  106400|  53200|    0|
+-----------------+---------+-----+--------+-------+-----+
|Utilization (%)  |        0|   15|       4|      5|    0|
+-----------------+---------+-----+--------+-------+-----+

+ Detail:
    * Instance:
    +----------------------------------------+-------------------------------------+---------+----+------+------+-----+
    |                Instance                |                Module               | BRAM_18K| DSP|  FF  |  LUT | URAM|
    +----------------------------------------+-------------------------------------+---------+----+------+------+-----+
    |control_s_axi_U                         |control_s_axi                        |        0|   0|   294|   436|    0|
    |grp_fir_n11_maxi_Pipeline_XFER_LOOP_fu_242 |fir_n11_maxi_Pipeline_XFER_LOOP   |        0|  33|  2794|  1084|    0|
    |gmem_m_axi_U                            |gmem_m_axi                           |        0|   0|   718|  1318|    0|
    +----------------------------------------+-------------------------------------+---------+----+------+------+-----+
    |Total                                   |                                     |        0|  33|  3806|  2838|    0|
    +----------------------------------------+-------------------------------------+---------+----+------+------+-----+

    * DSP:
    N/A

    * Memory:
    N/A

    * FIFO:
    N/A

    * Expression:
    +-------------------+----------+----+---+----+------------+------------+
    |   Variable Name   | Operation| DSP| FF| LUT| Bitwidth P0| Bitwidth P1|
    +-------------------+----------+----+---+----+------------+------------+
    |add_ln16_fu_289_p2 |         +|   0|  0|  40|          33|           2|
    +-------------------+----------+----+---+----+------------+------------+
    |Total              |          |   0|  0|  40|          33|           2|
    +-------------------+----------+----+---+----+------------+------------+

    * Multiplexer:
    +-------------------+----+-----------+-----+-----------+
    |        Name       | LUT| Input Size| Bits| Total Bits|
    +-------------------+----+-----------+-----+-----------+
    |an32Coef_address0  |  65|         12|    4|         48|
    |ap_NS_fsm          |  65|         15|    1|         15|
    |gmem_ARVALID       |   9|          2|    1|          2|
    |gmem_AWVALID       |   9|          2|    1|          2|
    |gmem_BREADY        |   9|          2|    1|          2|
    |gmem_RREADY        |   9|          2|    1|          2|
    |gmem_WVALID        |   9|          2|    1|          2|
    +-------------------+----+-----------+-----+-----------+
    |Total              | 175|         37|   10|         73|
    +-------------------+----+-----------+-----+-----------+

    * Register:
    +--------------------------------------------------------+----+----+-----+-----------+
    |                          Name                          | FF | LUT| Bits| Const Bits|
    +--------------------------------------------------------+----+----+-----+-----------+
    |an32Coef_load_10_reg_446                                |  32|   0|   32|          0|
    |an32Coef_load_1_reg_340                                 |  32|   0|   32|          0|
    |an32Coef_load_2_reg_350                                 |  32|   0|   32|          0|
    |an32Coef_load_3_reg_360                                 |  32|   0|   32|          0|
    |an32Coef_load_4_reg_370                                 |  32|   0|   32|          0|
    |an32Coef_load_5_reg_380                                 |  32|   0|   32|          0|
    |an32Coef_load_6_reg_390                                 |  32|   0|   32|          0|
    |an32Coef_load_7_reg_400                                 |  32|   0|   32|          0|
    |an32Coef_load_8_reg_410                                 |  32|   0|   32|          0|
    |an32Coef_load_9_reg_420                                 |  32|   0|   32|          0|
    |an32Coef_load_reg_330                                   |  32|   0|   32|          0|
    |ap_CS_fsm                                               |  14|   0|   14|          0|
    |grp_fir_n11_maxi_Pipeline_XFER_LOOP_fu_242_ap_start_reg |   1|   0|    1|          0|
    |lshr_ln16_cast_reg_440                                  |  31|   0|   31|          0|
    |pn32HPInput_read_reg_435                                |  64|   0|   64|          0|
    |pn32HPOutput_read_reg_430                               |  64|   0|   64|          0|
    |trunc_ln18_1_reg_451                                    |  62|   0|   62|          0|
    |trunc_ln30_1_reg_456                                    |  62|   0|   62|          0|
    +--------------------------------------------------------+----+----+-----+-----------+
    |Total                                                   | 650|   0|  650|          0|
    +--------------------------------------------------------+----+----+-----+-----------+
```

Utilization (AXI-Master)

```
================================================================
== Utilization Estimates
================================================================
* Summary:
+-----------------+---------+-----+--------+-------+-----+
|       Name      | BRAM_18K| DSP |   FF   |  LUT  | URAM|
+-----------------+---------+-----+--------+-------+-----+
|DSP              |        -|    -|       -|      -|    -|
|Expression       |        -|    -|       0|     42|    -|
|FIFO             |        -|    -|       -|      -|    -|
|Instance         |        0|   33|    2988|   1333|    -|
|Memory           |        -|    -|       -|      -|    -|
|Multiplexer      |        -|    -|       -|     34|    -|
|Register         |        -|    -|      36|      -|    -|
+-----------------+---------+-----+--------+-------+-----+
|Total            |        0|   33|    3024|   1409|    0|
+-----------------+---------+-----+--------+-------+-----+
|Available        |      280|  220|  106400|  53200|    0|
+-----------------+---------+-----+--------+-------+-----+
|Utilization (%)  |        0|   15|       2|      2|    0|
+-----------------+---------+-----+--------+-------+-----+

+ Detail:
    * Instance:
    +--------------------------------------------+----------------------------------+---------+----+------+------+-----+
    |                  Instance                  |              Module              | BRAM_18K| DSP|  FF  |  LUT | URAM|
    +--------------------------------------------+----------------------------------+---------+----+------+------+-----+
    |control_s_axi_U                             |control_s_axi                     |        0|   0|   154|   180|    0|
    |grp_fir_n11_strm_Pipeline_XFER_LOOP_fu_112  |fir_n11_strm_Pipeline_XFER_LOOP   |        0|  33|  2834|  1153|    0|
    +--------------------------------------------+----------------------------------+---------+----+------+------+-----+
    |Total                                       |                                  |        0|  33|  2988|  1333|    0|
    +--------------------------------------------+----------------------------------+---------+----+------+------+-----+

    * DSP:
    N/A

    * Memory:
    N/A

    * FIFO:
    N/A

    * Expression:
    +-----------------------------------------------------------+----------+----+---+----+------------+------------+
    |                       Variable Name                       | Operation| DSP| FF| LUT| Bitwidth P0| Bitwidth P1|
    +-----------------------------------------------------------+----------+----+---+----+------------+------------+
    |ret_V_fu_171_p2                                            |         +|   0|  0|  40|          33|           2|
    |grp_fir_n11_strm_Pipeline_XFER_LOOP_fu_112_pstrmOutput_TREADY  |       and|   0|  0|   2|           1|           1|
    +-----------------------------------------------------------+----------+----+---+----+------------+------------+
    |Total                                                      |          |   0|  0|  42|          34|           3|
    +-----------------------------------------------------------+----------+----+---+----+------------+------------+

    * Multiplexer:
    +------------------------------+----+-----------+-----+-----------+
    |             Name             | LUT| Input Size| Bits| Total Bits|
    +------------------------------+----+-----------+-----+-----------+
    |ap_NS_fsm                     |  25|          5|    1|          5|
    |pstrmInput_TREADY_int_regslice  |   9|          2|    1|          2|
    +------------------------------+----+-----------+-----+-----------+
    |Total                         |  34|          7|    2|          7|
    +------------------------------+----+-----------+-----+-----------+

    * Register:
    +---------------------------------------------------+----+----+-----+-----------+
    |                        Name                       | FF | LUT| Bits| Const Bits|
    +---------------------------------------------------+----+----+-----+-----------+
    |ap_CS_fsm                                          |   4|   0|    4|          0|
    |grp_fir_n11_strm_Pipeline_XFER_LOOP_fu_112_ap_start_reg  |   1|   0|    1|          0|
    |tmp_reg_187                                        |  31|   0|   31|          0|
    +---------------------------------------------------+----+----+-----+-----------+
    |Total                                              |  36|   0|   36|          0|
    +---------------------------------------------------+----+----+-----+-----------+
```

Utilization (AXI-Stream)

- Interface

```
================================================================
== Interface
================================================================
* Summary:
+----------------------+-----+-----+-----------+-------------+-------------+
|       RTL Ports      | Dir | Bits|  Protocol | Source Object|    C Type   |
+----------------------+-----+-----+-----------+-------------+-------------+
|s_axi_control_AWVALID |   in|    1|      s_axi|      control|        array|
|s_axi_control_AWREADY |  out|    1|      s_axi|      control|        array|
|s_axi_control_AWADDR  |   in|    7|      s_axi|      control|        array|
|s_axi_control_WVALID  |   in|    1|      s_axi|      control|        array|
|s_axi_control_WREADY  |  out|    1|      s_axi|      control|        array|
|s_axi_control_WDATA   |   in|   32|      s_axi|      control|        array|
|s_axi_control_WSTRB   |   in|    4|      s_axi|      control|        array|
|s_axi_control_ARVALID |   in|    1|      s_axi|      control|        array|
|s_axi_control_ARREADY |  out|    1|      s_axi|      control|        array|
|s_axi_control_ARADDR  |   in|    7|      s_axi|      control|        array|
|s_axi_control_RVALID  |  out|    1|      s_axi|      control|        array|
|s_axi_control_RREADY  |   in|    1|      s_axi|      control|        array|
|s_axi_control_RDATA   |  out|   32|      s_axi|      control|        array|
|s_axi_control_RRESP   |  out|    2|      s_axi|      control|        array|
|s_axi_control_BVALID  |  out|    1|      s_axi|      control|        array|
|s_axi_control_BREADY  |   in|    1|      s_axi|      control|        array|
|s_axi_control_BRESP   |  out|    2|      s_axi|      control|        array|
|ap_clk                |   in|    1| ap_ctrl_hs| fir_n11_maxi| return value|
|ap_rst_n              |   in|    1| ap_ctrl_hs| fir_n11_maxi| return value|
|interrupt             |  out|    1| ap_ctrl_hs| fir_n11_maxi| return value|
|m_axi_gmem_AWVALID    |  out|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_AWREADY    |   in|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_AWADDR     |  out|   64|      m_axi|         gmem|      pointer|
|m_axi_gmem_AWID       |  out|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_AWLEN      |  out|    8|      m_axi|         gmem|      pointer|
|m_axi_gmem_AWSIZE     |  out|    3|      m_axi|         gmem|      pointer|
|m_axi_gmem_AWBURST    |  out|    2|      m_axi|         gmem|      pointer|
|m_axi_gmem_AWLOCK     |  out|    2|      m_axi|         gmem|      pointer|
|m_axi_gmem_AWCACHE    |  out|    4|      m_axi|         gmem|      pointer|
|m_axi_gmem_AWPROT     |  out|    3|      m_axi|         gmem|      pointer|
|m_axi_gmem_AWQOS      |  out|    4|      m_axi|         gmem|      pointer|
|m_axi_gmem_AWREGION   |  out|    4|      m_axi|         gmem|      pointer|
|m_axi_gmem_AWUSER     |  out|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_WVALID     |  out|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_WREADY     |   in|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_WDATA      |  out|   32|      m_axi|         gmem|      pointer|
|m_axi_gmem_WSTRB      |  out|    4|      m_axi|         gmem|      pointer|
|m_axi_gmem_WLAST      |  out|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_WID        |  out|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_WUSER      |  out|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_ARVALID    |  out|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_ARREADY    |   in|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_ARADDR     |  out|   64|      m_axi|         gmem|      pointer|
|m_axi_gmem_ARID       |  out|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_ARLEN      |  out|    8|      m_axi|         gmem|      pointer|
|m_axi_gmem_ARSIZE     |  out|    3|      m_axi|         gmem|      pointer|
|m_axi_gmem_ARBURST    |  out|    2|      m_axi|         gmem|      pointer|
|m_axi_gmem_ARLOCK     |  out|    2|      m_axi|         gmem|      pointer|
|m_axi_gmem_ARCACHE    |  out|    4|      m_axi|         gmem|      pointer|
|m_axi_gmem_ARPROT     |  out|    3|      m_axi|         gmem|      pointer|
|m_axi_gmem_ARQOS      |  out|    4|      m_axi|         gmem|      pointer|
|m_axi_gmem_ARREGION   |  out|    4|      m_axi|         gmem|      pointer|
|m_axi_gmem_ARUSER     |  out|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_RVALID     |   in|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_RREADY     |  out|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_RDATA      |   in|   32|      m_axi|         gmem|      pointer|
|m_axi_gmem_RLAST      |   in|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_RID        |   in|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_RUSER      |   in|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_RRESP      |   in|    2|      m_axi|         gmem|      pointer|
|m_axi_gmem_BVALID     |   in|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_BREADY     |  out|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_BRESP      |   in|    2|      m_axi|         gmem|      pointer|
|m_axi_gmem_BID        |   in|    1|      m_axi|         gmem|      pointer|
|m_axi_gmem_BUSER      |   in|    1|      m_axi|         gmem|      pointer|
+----------------------+-----+-----+-----------+-------------+-------------+
```

Interface (AXI-Master)

```
================================================================
== Interface
================================================================
* Summary:
+------------------------+-----+-----+------------+--------------------+--------------+
|        RTL Ports       | Dir | Bits|  Protocol  |    Source Object   |    C Type    |
+------------------------+-----+-----+------------+--------------------+--------------+
|s_axi_control_AWVALID   |   in|    1|       s_axi|             control|         array|
|s_axi_control_AWREADY   |  out|    1|       s_axi|             control|         array|
|s_axi_control_AWADDR    |   in|    7|       s_axi|             control|         array|
|s_axi_control_WVALID    |   in|    1|       s_axi|             control|         array|
|s_axi_control_WREADY    |  out|    1|       s_axi|             control|         array|
|s_axi_control_WDATA     |   in|   32|       s_axi|             control|         array|
|s_axi_control_WSTRB     |   in|    4|       s_axi|             control|         array|
|s_axi_control_ARVALID   |   in|    1|       s_axi|             control|         array|
|s_axi_control_ARREADY   |  out|    1|       s_axi|             control|         array|
|s_axi_control_ARADDR    |   in|    7|       s_axi|             control|         array|
|s_axi_control_RVALID    |  out|    1|       s_axi|             control|         array|
|s_axi_control_RREADY    |   in|    1|       s_axi|             control|         array|
|s_axi_control_RDATA     |  out|   32|       s_axi|             control|         array|
|s_axi_control_RRESP     |  out|    2|       s_axi|             control|         array|
|s_axi_control_BVALID    |  out|    1|       s_axi|             control|         array|
|s_axi_control_BREADY    |   in|    1|       s_axi|             control|         array|
|s_axi_control_BRESP     |  out|    2|       s_axi|             control|         array|
|ap_clk                  |   in|    1|  ap_ctrl_hs|        fir_n11_strm|  return value|
|ap_rst_n                |   in|    1|  ap_ctrl_hs|        fir_n11_strm|  return value|
|interrupt               |  out|    1|  ap_ctrl_hs|        fir_n11_strm|  return value|
|pstrmInput_TDATA        |   in|   32|        axis|   pstrmInput_V_data_V|       pointer|
|pstrmInput_TVALID       |   in|    1|        axis|   pstrmInput_V_dest_V|       pointer|
|pstrmInput_TREADY       |  out|    1|        axis|   pstrmInput_V_dest_V|       pointer|
|pstrmInput_TDEST        |   in|    1|        axis|   pstrmInput_V_dest_V|       pointer|
|pstrmInput_TKEEP        |   in|    4|        axis|   pstrmInput_V_keep_V|       pointer|
|pstrmInput_TSTRB        |   in|    4|        axis|   pstrmInput_V_strb_V|       pointer|
|pstrmInput_TUSER        |   in|    1|        axis|   pstrmInput_V_user_V|       pointer|
|pstrmInput_TLAST        |   in|    1|        axis|   pstrmInput_V_last_V|       pointer|
|pstrmInput_TID          |   in|    1|        axis|     pstrmInput_V_id_V|       pointer|
|pstrmOutput_TDATA       |  out|   32|        axis|  pstrmOutput_V_data_V|       pointer|
|pstrmOutput_TVALID      |  out|    1|        axis|  pstrmOutput_V_dest_V|       pointer|
|pstrmOutput_TREADY      |   in|    1|        axis|  pstrmOutput_V_dest_V|       pointer|
|pstrmOutput_TDEST       |  out|    1|        axis|  pstrmOutput_V_dest_V|       pointer|
|pstrmOutput_TKEEP       |  out|    4|        axis|  pstrmOutput_V_keep_V|       pointer|
|pstrmOutput_TSTRB       |  out|    4|        axis|  pstrmOutput_V_strb_V|       pointer|
|pstrmOutput_TUSER       |  out|    1|        axis|  pstrmOutput_V_user_V|       pointer|
|pstrmOutput_TLAST       |  out|    1|        axis|  pstrmOutput_V_last_V|       pointer|
|pstrmOutput_TID         |  out|    1|        axis|    pstrmOutput_V_id_V|       pointer|
+------------------------+-----+-----+------------+--------------------+--------------+
```

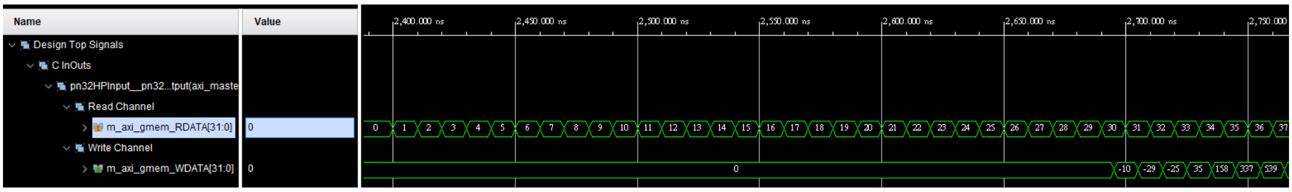
Interface (AXI-Stream)

- Co-simulation transcript/waveform

```
Report time       : Tue Sep 26 20:10:50     2023.
Solution          : solution1.
Simulation tool   : xsim.

+---------+---------+------------------------------------------------+------------------------------------------------+--------------------+
|         |         |             Latency(Clock Cycles)              |              Interval(Clock Cycles)            | Total Execution Time |
+   RTL   + Status  +------------------------------------------------+------------------------------------------------+    (Clock Cycles)  +
|         |         |      min      |      avg      |      max      |      min      |      avg      |      max      |                    |
+---------+---------+------------------------------------------------+------------------------------------------------+--------------------+
|     VHDL|       NA|             NA|             NA|             NA|             NA|             NA|             NA|                  NA|
|  Verilog|     Pass|            943|            943|            943|             NA|             NA|             NA|                 943|
+---------+---------+------------------------------------------------+------------------------------------------------+--------------------+
```

Co-simulation transcript (AXI-Master)



Co-simulation waveform (AXI-Master)

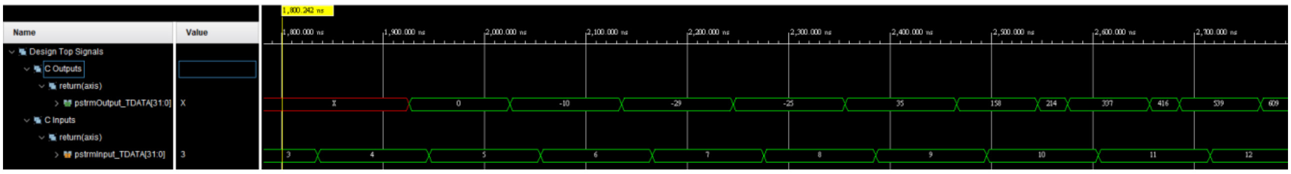
Index	Input	Output
0	1	0
1	2	-10
2	3	-29
3	4	-25
4	5	35
5	6	158
6	7	337
7	8	539

out_gold.dat

```
Report time      : Tue Sep 26 20:10:50    2023.
Solution         : solution1.
Simulation tool  : xsim.

+---------+---------+-----------------------------------------------+-----------------------------------------------+----------------------+
|         |         |             Latency(Clock Cycles)             |             Interval(Clock Cycles)            | Total Execution Time |
+   RTL   +  Status +-----------------------------------------------+-----------------------------------------------+    (Clock Cycles)    +
|         |         |     min     |     avg     |     max     |     min     |     avg     |     max     |                      |
+---------+---------+-----------------------------------------------+-----------------------------------------------+----------------------+
|     VHDL|       NA|           NA|           NA|           NA|           NA|           NA|           NA|                    NA|
|  Verilog|     Pass|          943|          943|          943|           NA|           NA|           NA|                   943|
+---------+---------+-----------------------------------------------+-----------------------------------------------+----------------------+
```

Co-simulation transcript (AXI-Stream)



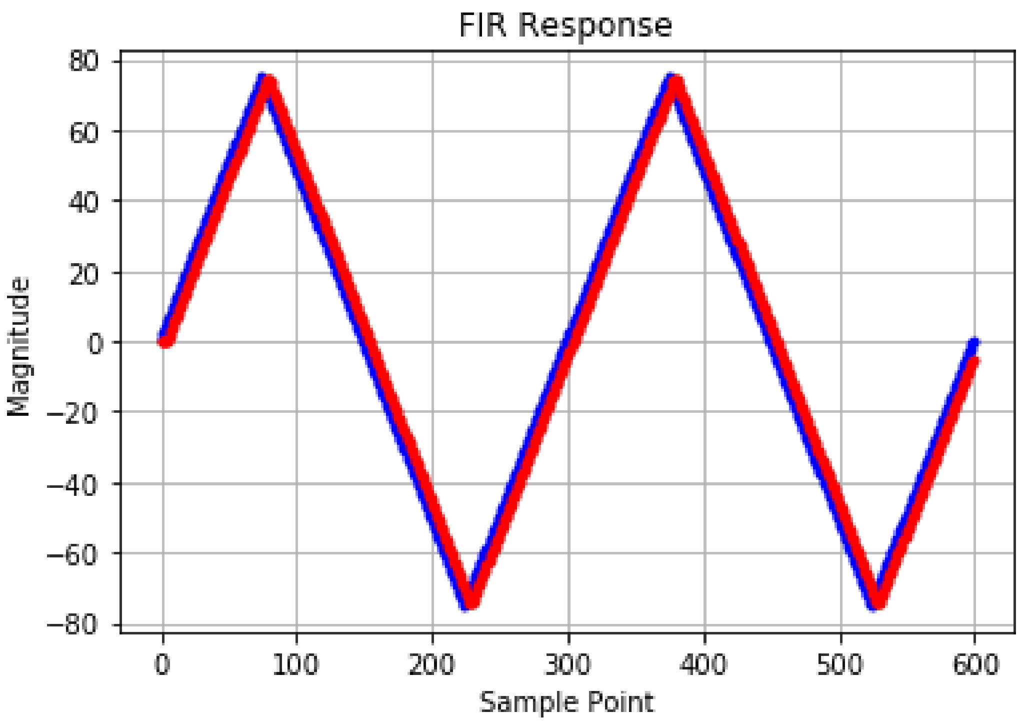
Co-simulation waveform (AXI-Stream)

Index	Input	Output
0	1	0
1	2	-10
2	3	-29
3	4	-25
4	5	35
5	6	158
6	7	337
7	8	539

out_gold.dat

- Jupyter Notebook execution results
  AXI-Master:

```
Entry: /usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py
System argument(s): 3
Start of "/usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py"
Kernel execution time: 0.0008635520935058594 s
```



AXI-Stream:

```
Entry: /usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py
System argument(s): 3
Start of "/usr/local/share/pynq-venv/lib/python3.8/site-packages/ipykernel_launcher.py"
Kernel execution time: 0.001735687255859375 s
```